SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – EXPERIMENTATION

Course Code:	ITA03	**Course Title:**	Mobile computing
CO Assessed:	CO2	**Assessment:**	Assessment Tool 2 – EXPERIMENTATION
Weightage:	30 % of CIA	**Total Marks:**	100 marks
CO2 Statement:	*Compare mobile communication technologies, including multiple access techniques and network evolution, based on performance and applications.*		
Student Name:	c sangita	**Reg. No.:**	192373059
Section / Batch:	Slot-a	**Date:**	15-10-2026

Code of Conduct

I C Sangita(192373059) (Name / Reg. No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____sangita.c_____

Instructions

- This test contains FIVE questions.
- Duration: 60 minutes (homework)

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Annexure A - QUESTIONS

1. Figma-Based Mobile IP Communication Workflow Analysis (BL4 – Analyze)

Design an interactive Figma prototype illustrating the complete workflow of Mobile IP communication when a Mobile Node moves from its Home Network to a Foreign Network. Your design should include the Home Agent (HA), Foreign Agent (FA), Correspondent Node (CN), Care-of Address (CoA), Registration Process, and Packet Tunneling.

Analyze the communication process by identifying:

- The purpose of each network component.
- Points where packet delay or packet loss may occur.
- The effect of node mobility on communication performance.
- Possible improvements to reduce latency.

Deliverables:

- Figma (.fig) design
- GitHub repository containing exported images/PDF and documentation (README).

2. Comparative Network Architecture Design Using Figma (BL4 – Analyze)

Using Figma or draw.io, create a comparative architectural diagram showing both a Cellular Wireless Network and a Mobile Ad Hoc Network (MANET) operating in the same smart city environment.

Analyze the architectures by comparing:

- Infrastructure dependency.
- Routing mechanisms.
- Scalability.
- Fault tolerance.
- Mobility support.
- Suitable application scenarios.

Highlight the advantages and limitations of each architecture through annotations in your design.

Deliverables:

- Editable design file.
- GitHub repository with screenshots, comparison report, and design assets.

3. Routing Protocol Visualization and Analysis (BL4 – Analyze)

Develop an interactive visualization using Figma, Mermaid.js, D3.js, or any visualization framework illustrating the operation of one Proactive Routing Protocol and one Reactive Routing Protocol in a mobile wireless network.

Analyze:

- Route discovery process.
- Route maintenance mechanism.
- Routing overhead.
- Delay in establishing communication.
- Performance under high node mobility.

Include a comparative summary explaining which protocol performs better under different mobility conditions.

Deliverables:

- Source code/design files.
- GitHub repository with implementation, screenshots, and analysis report.

4. Mobile IP Application Design for Smart Healthcare (BL4 – Analyze)

Design a Smart Healthcare Mobile IP system using Figma that enables doctors, ambulances, hospitals, and patients to remain connected while moving across different wireless networks.

Analyze:

- How Mobile IP maintains uninterrupted communication.
- Challenges during handoff between networks.
- Security vulnerabilities during mobility.
- Possible design improvements to increase reliability.

Provide navigation between screens demonstrating different mobility scenarios.

Deliverables:

- Interactive prototype.
- GitHub repository containing design files, exported prototype, and documentation.

5. Wireless Network Decision Dashboard Using GitHub Project (BL4 – Analyze)

Create a web-based dashboard (using HTML/CSS/JavaScript, React, or another programming framework) or an interactive Figma dashboard that helps network engineers determine whether a given scenario should use Mobile IP, Cellular Networks, or Ad Hoc Networks.

The dashboard should analyze multiple parameters such as:

- Node mobility.
- Infrastructure availability.
- Network size.
- Routing complexity.
- Energy consumption.
- Reliability requirements.

The system should justify its recommendation based on the analyzed parameters.

Deliverables:

- Source code or Figma project.
- GitHub repository with complete code/design, README, screenshots, and analysis report.

Criterion	Marks
Design Quality (Figma/Programming Tool)	20
Technical Analysis	30
Correctness of Mobile Networking Concepts	20
GitHub Repository Organization & Documentation	15
Creativity and Visualization	15
Total	**100 Marks**

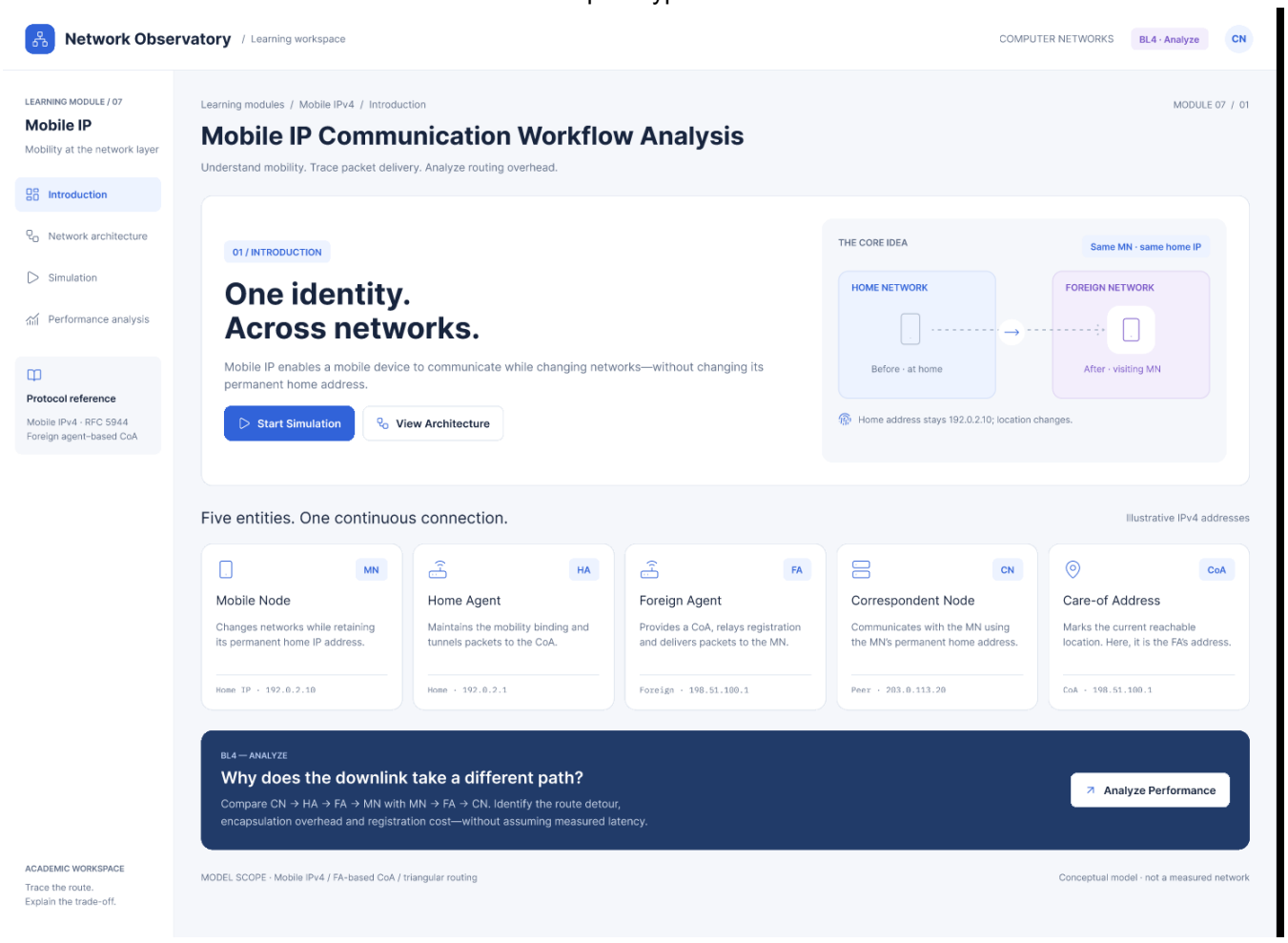
1. Figma-Based Mobile IP Communication Workflow Analysis (BL4 – Analyze)

Aim

To design an interactive Figma prototype illustrating Mobile IP communication when a Mobile Node moves from its Home Network to a Foreign Network, and to analyze packet delay, packet loss, mobility impact, and latency reduction techniques.

Algorithm

1. Open Figma and create a new design project.
2. Design the Home Network and Foreign Network with the MN, HA, FA, and CN.
3. Show the Care-of Address (CoA) when the MN moves to the Foreign Network.
4. Illustrate the registration request and reply between the MN and HA.
5. Demonstrate packet tunneling from the HA to the CoA and delivery to the MN.
6. Add interactive buttons to simulate each communication stage.
7. Identify possible packet delay and packet loss points.
8. Display techniques to reduce latency and improve communication performance.
9. Connect all screens and test the interactive prototype.



Mobile IP Communication Workflow Analysis

Understand mobility. Trace packet delivery. Analyze routing overhead.

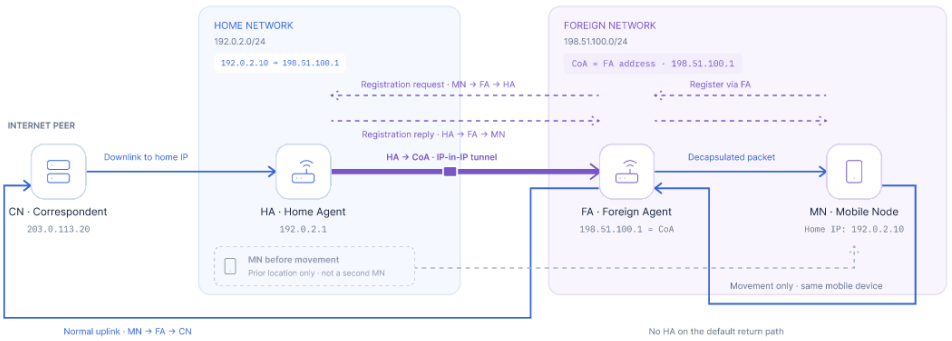
SCENARIO 01 **Home → Foreign network** / After registration

Reset Step Run Simulation

Network architecture

After movement · MN registered on the foreign network

Binding active



LEGEND — Normal packets — Tunneled packets ··· Registration messages ··· Device movement (not traffic)

Why is the CoA on the FA? In this FA-based model, the tunnel ends at the FA—not at the MN. The MN keeps 192.0.2.10.

Communication workflow

01 **Discover & register**
MN learns the FA's CoA; registration travels MN → FA → HA.

02 **Bind & acknowledge**
HA binds the home IP to the CoA; the reply returns via FA.

03 **Tunnel & deliver**

Route path & overhead

BL4 · Analyze

↓ Downlink CN → HA → FA → MN
↑ Uplink MN → FA → CN

Route detour · The HA anchor makes routing asymmetric. Extra path length depends on the nodes' locations—not a fixed delay.

2. Comparative Network Architecture Design Using Figma (BL4 – Analyze)

Aim

To design a comparative network architecture diagram using Figma, illustrating Cellular Wireless Networks and Mobile Ad Hoc Networks (MANETs) in a smart city and analyzing their infrastructure, routing, scalability, fault tolerance, mobility support, and applications.

Algorithm

1. Open Figma and create a new design project.
2. Draw a smart city environment with buildings, roads, and mobile devices.
3. Design the Cellular Wireless Network with mobile devices, base stations, a core network, and internet connectivity.
4. Design the MANET with mobile nodes communicating directly and through intermediate nodes.
5. Show the routing mechanisms and communication paths of both networks.
6. Add annotations describing the advantages and limitations of each architecture.
7. Create a comparison table covering infrastructure, routing, scalability, fault tolerance, mobility, and applications.
8. Add colors, labels, and legends to distinguish both network architectures.
9. Review the diagram and save the final design.

Comparative Network Architecture: Cellular Wireless Network vs Mobile Ad Hoc Network (MANET)

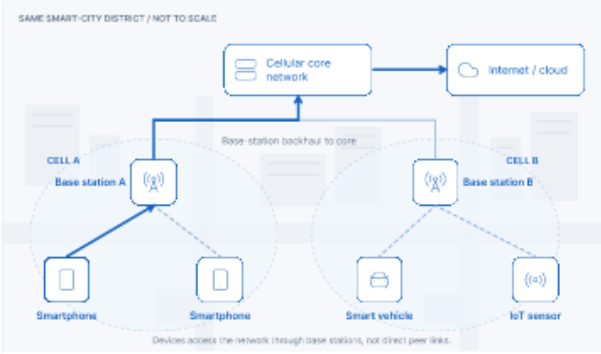
One smart-city district, two communication models: infrastructure-managed access versus self-organizing mobile peers.

01 Architecture & communication paths

Blue = cellular / Purple = MANET

Cellular Wireless Network

Infrastructure-based • Managed access • Centralized core

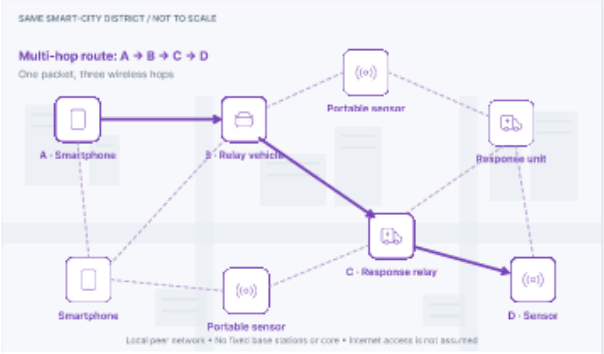


Handover example / Cell A → Cell B

A moving smartphone switches from base station A to base station B. The network coordinates handover to maintain the session.

Mobile Ad Hoc Network (MANET)

Distributed • Peer-to-peer • Infrastructure-independent



Intermediate-node forwarding / A → B → C → D

B and C relay source A's packet to destination D. Each hop depends on a nearby peer; A need not reach D directly.

READING THE DIAGRAMS — Dashed: available wireless link · Thin solid: cellular backhaul / core link · Thick arrow: example packet direction

Unarrowed links show connectivity; arrows trace one example packet. Return traffic may flow in reverse. Shaded ovals show illustrative cell coverage.

02 Compare the operating characteristics

Seven aligned comparison categories

COMPARISON CATEGORY	Cellular Wireless Network	Mobile Ad Hoc Network (MANET)
Infrastructure Dependency	Requires base stations and cellular infrastructure.	No fixed infrastructure; mobile devices form and maintain the network themselves.
Routing	Communication is managed through the cellular network infrastructure.	Distributed peer-to-peer routing with multi-hop forwarding through intermediate mobile nodes.
Scalability	Can support many users with sufficient network capacity and infrastructure.	Constrained by shared-channel contention, routing overhead, and dynamic topology as the network grows.
Fault Tolerance	Base station or core network failures may disrupt connectivity in affected areas.	Alternate routes can improve resilience where available, but disconnected network partitions are possible.
Mobility Support	Supports mobility through handover between cells.	Mobility changes the topology, requiring route discovery and repair as peer links appear or break.
Advantages	Wide-area coverage, managed connectivity, and centralized network control.	Rapid deployment, flexibility, and useful local emergency connectivity without fixed infrastructure.
Limitations	Infrastructure cost, coverage gaps, and possible congestion.	Limited range, battery, and bandwidth; security challenges and routing overhead.

03 BL4–Analyze: choose by operating conditions

Dense, routine smart-city use

Commuters, connected vehicles, and IoT sensors share a busy district.

Cellular: Managed capacity and handover suit large-scale daily connectivity, provided coverage and infrastructure capacity are sufficient.

MANET: Local peer exchange is possible, but contention, control traffic, and changing routes can limit performance as density grows.

Emergency after an infrastructure outage

Responders and portable sensors must communicate near an incident.

Cellular: Failed base stations or core links can interrupt service; recovery depends on surviving coverage or restored infrastructure.

MANET: Can form local multi-hop links rapidly, but needs reachable peers and power. Sparse nodes may partition; internet is not assured.

TRADEOFF, NOT A WINNER — Cellular trades infrastructure dependence for coordinated scale and mobility. MANET trades that infrastructure for rapid local deployment, with resource and topology constraints.

Conceptual comparison • Actual performance depends on deployment, radio conditions, and routing protocol.

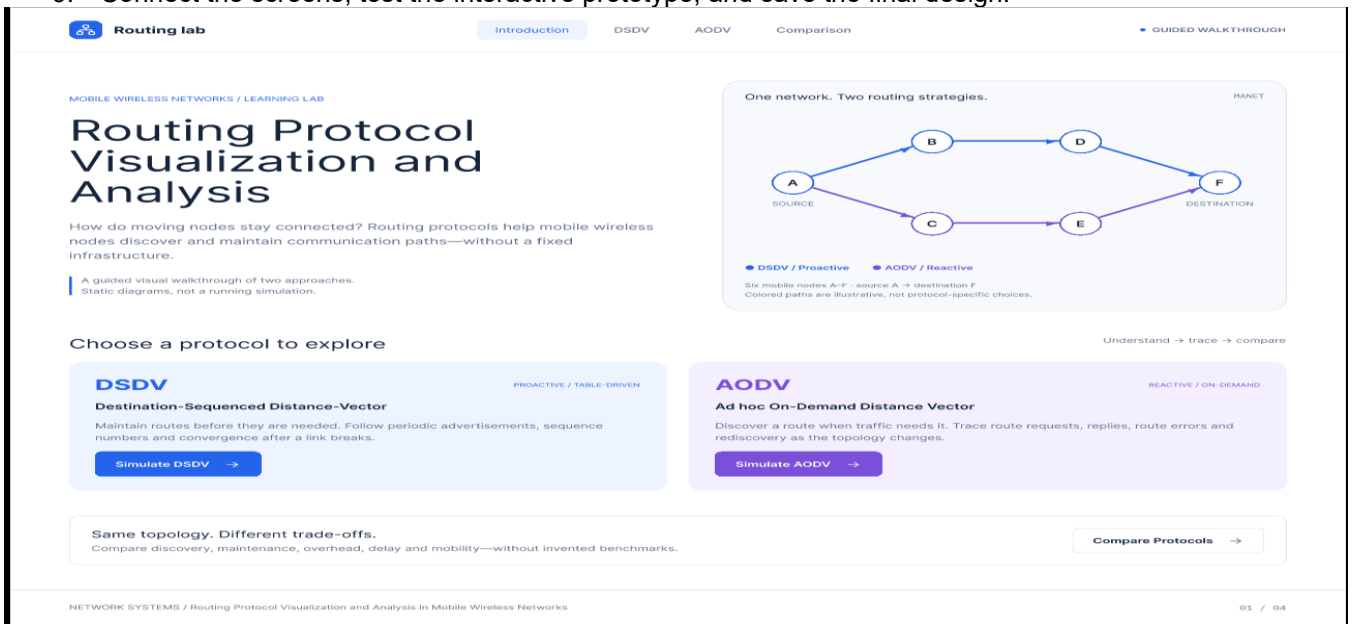
3.Routing Protocol Visualization and Analysis :

Aim

To develop an interactive visualization comparing the proactive routing protocol DSDV and the reactive routing protocol AODV in a mobile wireless network, and to analyze their routing overhead, delay, route maintenance, and performance under different mobility conditions.

Algorithm

1. Open Figma and create a new design project.
2. Draw a mobile wireless network with multiple interconnected nodes.
3. Illustrate DSDV routing using periodic routing table updates.
4. Illustrate AODV routing using RREQ, RREP, and RERR messages.
5. Simulate packet transmission and route establishment for both protocols.
6. Show link failures and route maintenance when nodes move.
7. Analyze routing overhead, communication delay, and performance under different mobility conditions.
8. Create a comparison table highlighting the advantages and limitations of both protocols.
9. Connect the screens, test the interactive prototype, and save the final design.



02 / MOBILE WIRELESS NETWORKS

Static learning sequence · not a running simulation

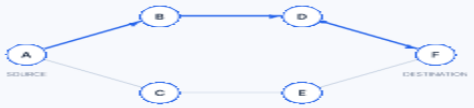
DSDV: maintain first, transmit when ready.

Destination-Sequenced Distance-Vector · A proactive, table-driven protocol that distributes routes before data is sent.

Before mobility - route ready

STAGES 03–04

Data follows the converged route A → B → D → F.



Wireless link — Selected route →

● Small ellipse on A–B = data packet

After mobility - route recovered

STAGES 05–06

B–D breaks. After convergence: A → C → E → F.



Wireless link — Selected route → — Failed link ×

D moves out of B's range; B–F remains available.

Routing table at A

Converged · before mobility

Destination	Next hop	Hop count	Dest. sequence no.
A	—	0	20
B	B	1	30
C	C	1	24
D	B	2	38
E	C	2	26
F	B	3	40

Freshness before distance.

A newer destination sequence number wins, even if the path is longer. Hop count breaks ties only between equally fresh routes.

F: via B · 3 hops · seq 40
→ invalid · ∞ hops · seq 41
→ via C · 3 hops · seq 42

Illustrative routing state, not measured performance. The fresh valid advertisement supersedes the failure entry.

Follow the protocol, step by step

06 stages · read in order

01 Initialize routing tables

Each node records destination, next hop, hop count and destination sequence number. Start with the self route; discover neighbors. Unknown destinations have an infinite metric.

At A: A / – / 0 / 20; F / – / ∞ / –

02 Broadcast to neighbors

Nodes periodically advertise routing information. Full dumps carry the whole table; incremental updates carry changes. Neighbors combine advertisements with their own tables.

Control updates continue even without data traffic.

03 Learn routes before traffic

Prefer the fresher destination sequence number; only when freshness is equal, prefer fewer hops. After convergence, A already has a valid three-hop route to F through B.

Compare sequence freshness first, then hop count.

04 Forward the data packet

The packet follows A → B → D → F. Each node forwards using its own next-hop entry. No route-discovery exchange is needed for this valid, converged route.

Ready routes give low setup delay after convergence.

05 Detect a broken link

D moves out of B's range; B–D fails. Routes using that link become invalid. B advertises F with an infinite metric and an odd sequence number to mark the failure.

Example invalidation: F / – / ∞ / 41

06 Trigger updates and converge

Triggered advertisements spread the change. A accepts a fresh sequence 42 route via C, E and F. Once updates converge, data can use the valid alternate path.

Recovered at A: F / C / 3 / 42

Previous step → Next step → ↻ Restart

Visual controls only · all six stages are shown above

03 / MOBILE WIRELESS NETWORKS

Static learning sequence · not a running simulation

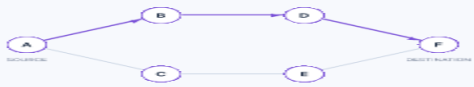
AODV: discover only when there is demand.

Ad hoc On-Demand Distance Vector · A reactive protocol that creates a route when a valid one is needed for traffic.

Before mobility - route discovered

STAGES 02–04

Only after RREQ / RREP: data A → B → D → F.



Wireless link — Selected route →

Initially, no valid A → F route exists.

After mobility - route rediscovered

STAGES 05–06

After RERR and a new discovery: A → C → E → F.



Wireless link — Selected route → — Failed link ×

Same B–D failure as DSDV; D–F is still available.

Discovery is a conversation

RREQ — A → B, C → D, E → F
Broadcast outward; duplicates suppressed (A, 7) copies.

REVERSE — F → D → B → A
Accepted request branch provides the return path.

RREP — F → D → B → A
Unicast reply installs forward next hops toward F.

B–D is the accepted request branch in this example; another request arrival order may choose a different valid path.

A known route is not always a valid route.

Use destination sequence numbers for freshness and route lifetimes for validity. An expired or invalid entry cannot simply forward the next packet.

DEST / NEXT HOP / HOPS / SEQ / STATE
F / B / 3 / 40 / active
At A after RREP · lifetime unexpired

No valid cache → wait for discovery. Valid cache → avoid that initial discovery exchange.

Follow the protocol, step by step

06 stages · read in order

01 Start with no valid route

A has data for F but no valid routing entry. It cannot send yet. A starts discovery; if a cached route were still valid and unexpired, it could use that route instead.

At A: F / – / unknown / invalid

02 Flood the route request

A broadcasts an RREQ. Receivers establish reverse routes toward A and rebroadcast a new request. Duplicate copies with the same originator and RREQ ID are discarded.

Duplicate key: (originator A, RREQ ID 7)

03 Unicast the route reply

F sends an RREP back via D and B along the reverse route. A sufficiently fresh, valid intermediate route may also reply when permitted. Reply processing installs forward routes to F.

RREP: F → D → B → A; compare dest. sequence.

04 Send over the active route

After the reply reaches A, data follows A → B → D → F. Route lifetimes govern validity; active use refreshes relevant entries. The first transmission waited for discovery.

Data: A → B → D → F; route active, lifetime valid.

05 Detect failure and send RERR

When D moves and B–D fails, affected routes are invalidated. A route error (RERR) notifies upstream precursors. Local repair may be possible; it is not used in this example.

RERR: B → A; route to F marked invalid.

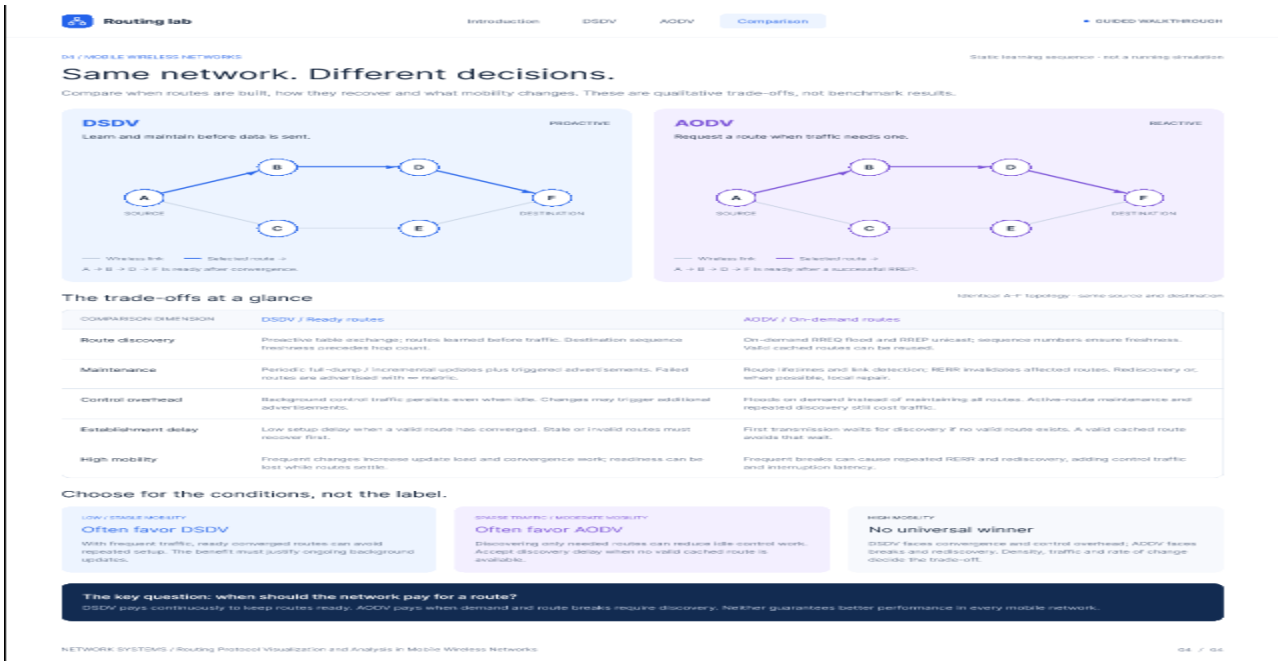
06 Rediscover an alternate path

If traffic continues, A sends a new RREQ with a new ID and updated freshness requirement. A fresh RREP establishes A → C → E → F; data resumes on the valid route.

New request ID 8 → fresh reply → forward data.

Previous step → Next step → ↻ Restart

Visual controls only · all six stages are shown above



4.Mobile IP Application Design for Smart Healthcare

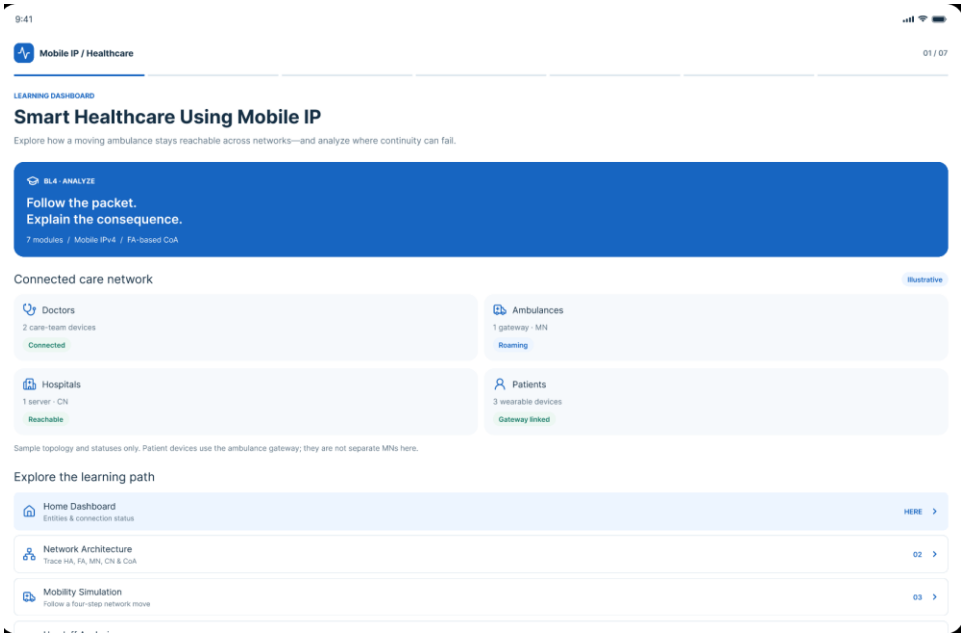
Aim

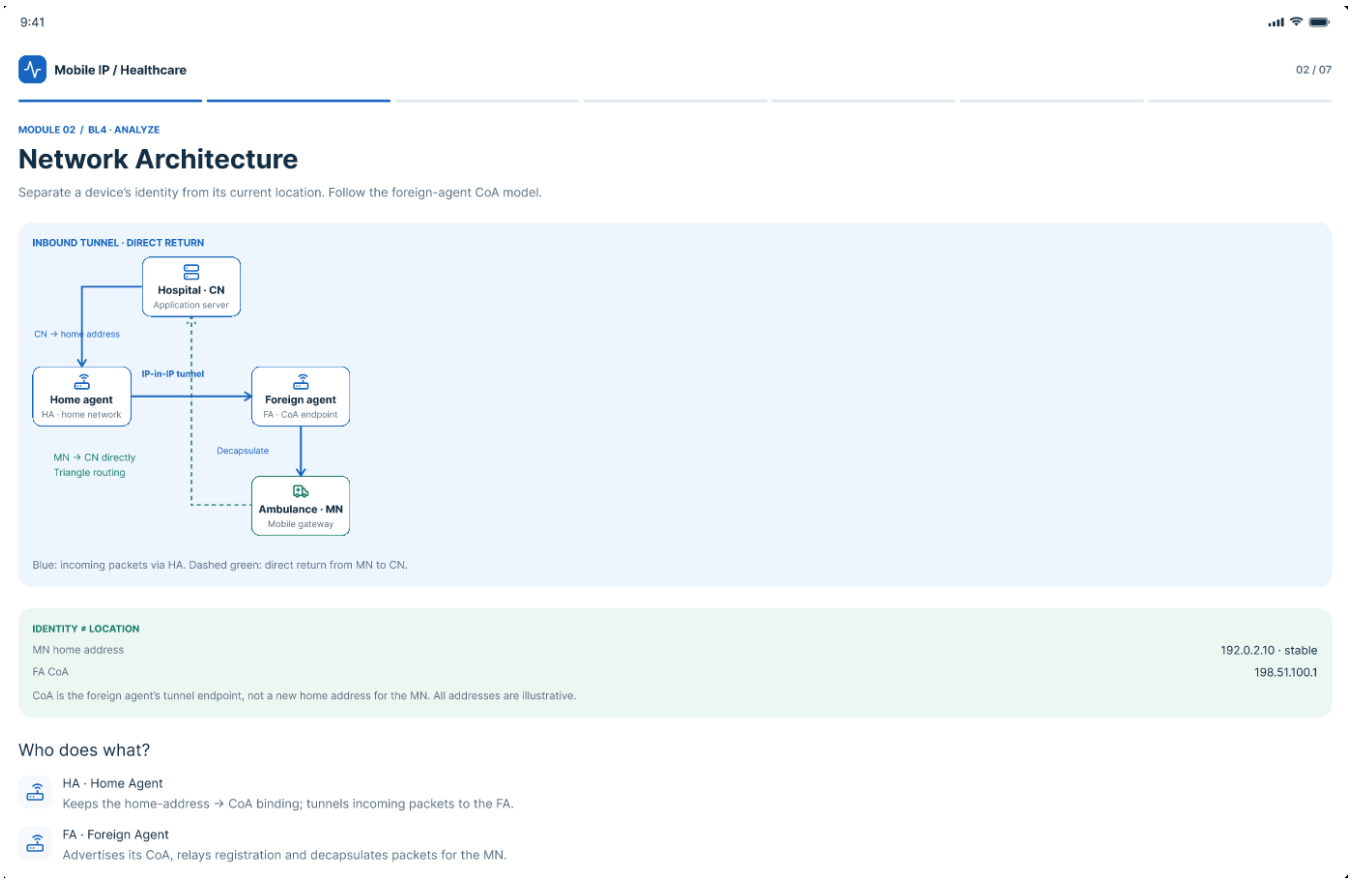
To design a Smart Healthcare Mobile IP system using Figma that maintains communication between doctors, ambulances, hospitals, and patients while moving across wireless networks, and to analyze handoff challenges, security vulnerabilities, and reliability improvements.

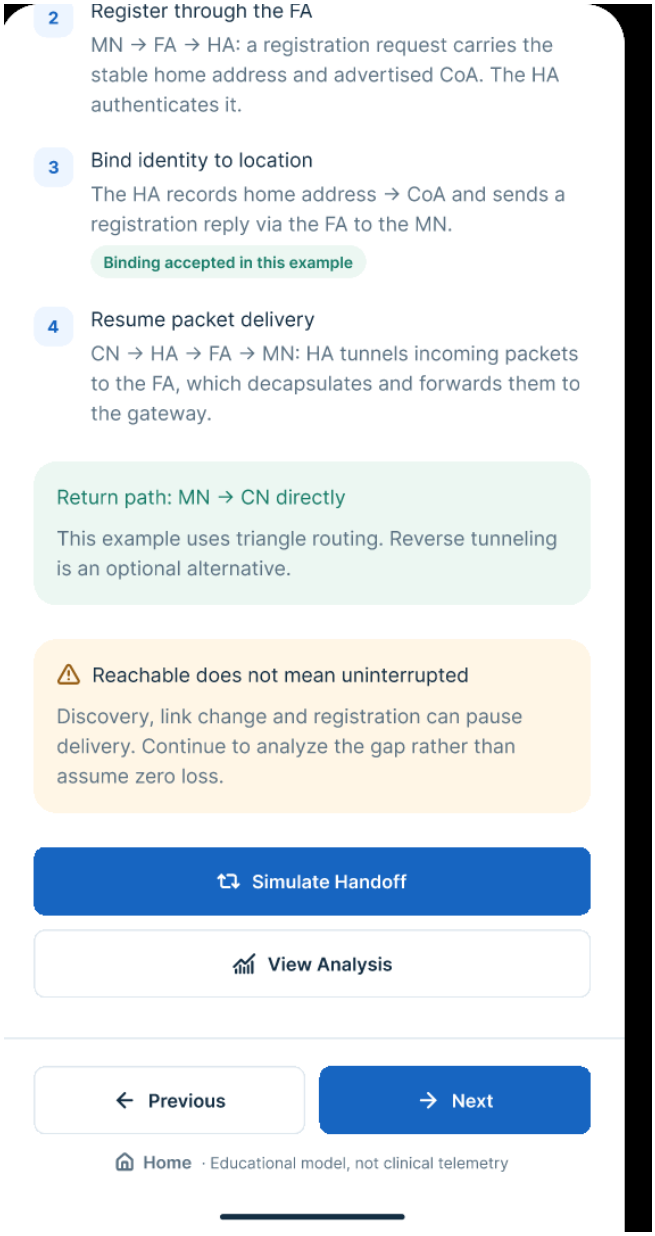
Algorithm

1. Open Figma and create a new design project.
2. Design a home dashboard showing doctors, ambulances, hospitals, and patients.
3. Create a network architecture containing MN, HA, FA, CN, and CoA.
4. Illustrate ambulance mobility between home and foreign networks.
5. Demonstrate registration, packet tunneling, and continued communication.
6. Identify handoff delays, packet loss, and security vulnerabilities.
7. Add solutions such as authentication, encryption, fast handoff, and packet buffering.
8. Connect all screens using interactive navigation buttons.
9. Test the prototype and save the final design.

10.


9:41
Mobile IP / Healthcare
01 / 07
LEARNING DASHBOARD
Smart Healthcare Using Mobile IP
Explore how a moving ambulance stays reachable across networks—and analyze where continuity can fail.
BL4 · ANALYZE
Follow the packet.
Explain the consequence.
7 modules / Mobile IPv4 / FA-based CoA
Connected care network
Illustrative
Doctors
2 care-team devices
Connected
Ambulances
1 gateway · MN
Roaming
Hospitals
1 server · CN
Reachable
Patients
3 wearable devices
Gateway linked
Sample topology and statuses only. Patient devices use the ambulance gateway; they are not separate MNs here.
Explore the learning path
Home Dashboard
Entities & connection status
HERE
Network Architecture
Trace HA, FA, MN, CN & CoA
02
Mobility Simulation
Follow a four-step network move
03


9:41
Mobile IP / Healthcare
02 / 07
MODULE 02 / BL4 · ANALYZE
Network Architecture
Separate a device's identity from its current location. Follow the foreign-agent CoA model.
INBOUND TUNNEL · DIRECT RETURN
Hospital · CN
Application server
CN → home address
IP-in-IP tunnel
Home agent
HA · home network
Foreign agent
FA · CoA endpoint
MN → CN directly
Triangle routing
Decapsulate
Ambulance · MN
Mobile gateway
Blue: incoming packets via HA. Dashed green: direct return from MN to CN.
IDENTITY ≠ LOCATION
MN home address
192.0.2.10 · stable
FA CoA
198.51.100.1
CoA is the foreign agent's tunnel endpoint, not a new home address for the MN. All addresses are illustrative.
Who does what?
HA · Home Agent
Keeps the home-address → CoA binding; tunnels incoming packets to the FA.
FA · Foreign Agent
Advertises its CoA, relays registration and decapsulates packets for the MN.



5.Wireless Network Decision Dashboard Using GitHub Project

Aim

To design an interactive Wireless Network Decision Dashboard using Figma that recommends Mobile IP, Cellular Networks, or Ad Hoc Networks based on node mobility, infrastructure availability, network size, routing complexity, energy consumption, and reliability requirements.

Algorithm

1. Open Figma and create a new design project.
2. Design a dashboard with three network options: Mobile IP, Cellular Network, and Ad Hoc Network.
3. Create input fields for mobility, infrastructure availability, network size, routing complexity, energy consumption, and reliability.
4. Add an Analyze button to evaluate the selected parameters.
5. Compare the network types based on the given scenario.
6. Display the recommended network architecture with suitable reasons.

7. Show the advantages, limitations, and comparison chart.
8. Connect all screens using interactive navigation buttons.
9. Test the prototype and save the final design.

